

KEVIN EPPACHER

Robotics Software Engineer

@ kevin-eppacher@hotmail.de
in linkedin.com/in/kevin-eppacher

+43 681 10499133
github.com/KevinEppacher

Vienna, Austria

kevin-eppacher.github.io



EXPERIENCE

Robotics Software Engineer

Baubot GmbH

Mar 2026 – Sep 2026

Vienna, Austria

- Owned the development of the ROS 2 software stack from scratch.
- Developed custom C++ action APIs for navigation and motion planning based on system-specific requirements.
- Built automated unit tests, headless Gazebo integration tests, and CI/CD pipelines using GitHub Actions.
- Developed a Python-based navigation controller using Regulated Pure Pursuit on the physical robot.
- Deployed and validated the ROS 2 stack on industrial PC hardware.

Research Assistant – Tutorial Author

UAS Technikum Vienna

Aug 2025 – Oct 2025

Vienna, Austria

- Authored student tutorials on ROS 2, nonlinear optimization with PyTorch, and end-to-end deep reinforcement learning using Stable-Baselines and Gymnasium.

Industrial Robotics Software Engineer

Blue Danube Robotics GmbH (AIRSKIN)

Aug 2023 – Jul 2025

Vienna, Austria

- Developed and deployed low-level robot programs (MoveIt C++ / KRL / RAPID) involving kinematics, tool calibration, coordinate-frame transformations, and precise motion execution for AIRSKIN-equipped robots.
- Built an automated sensitivity measurement system, including depth-based visualization of contact points and geometric evaluation of measurement data.
- Created integration concepts for customer-specific AIRSKIN solutions, including cell layout design, workspace modeling, and feasibility analysis.
- Supported quality inspection workflows and robotic handling tasks with geometry-aware motion planning and cycle-time optimization.

Bachelor Thesis Intern

Automation and Control Institute (ACIN), TU Vienna

Feb 2023 – Jul 2023

Vienna, Austria

- Developed a control system for high-speed drone tracking using a pan-tilt camera and cascaded PID controller.
- Implemented Kalman-filter-based sensor fusion for camera-based tracking and trajectory prediction in ROS / OpenCV.

EDUCATION

PROFILE

Robotics Software Engineer specializing in autonomous mobile robot navigation, motion planning, control, and ROS 2 system architecture. Experienced in developing production-oriented robotics software in C++ and Python, with hands-on experience in navigation and manipulation on physical robots.

SKILLS

Programming Languages

C/C++

Python

Frameworks & Libraries

ROS 2

Nav2 / MoveIt 2

PyTorch / CUDA

Gazebo

OpenCV

BehaviorTree.CPP

Gymnasium (RL)

Kuka AMR/KMP

Tools & Environment

Git / GitHub

CI/CD / GitHub Actions

Docker / Podman

VS Code

CMake

Linux

Isaac Sim

Computer Vision & AI

Tracking & 3D Geometry

Visual-Language Models (VLM)

Object Detection & Segmentation

Robotics & Control

Autonomous Navigation

Motion Planning

Localization / SLAM

State Estimation

Model Predictive Control

Nonlinear Optimization

Kinematics & Dynamics

M.Sc. Robotics

UAS Technikum Vienna

📅 2023 – 2026

📍 Vienna, Austria

- Focus on computer vision, probabilistic robotics, and nonlinear optimization.
- Thesis: “SAGE – Semantic-Aware Guided Exploration with Persistent Memory”.

B.Sc. Mechatronics & Robotics

UAS Technikum Vienna

📅 2020 – 2023

📍 Vienna, Austria

- Focus on dynamics, control theory, and mobile and articulated robotics.
- Thesis: “Design of a Cascaded Position and Velocity Controller for a Pan-Tilt Camera Tracking UAVs.”

Higher Technical College (HTL) – Mechatronics

HTL Mödling

📅 2015 – 2020

📍 Mödling, Austria

- Specialized in control theory, mechanical design, and electrical engineering.
- Graduated with a diploma in Mechatronics Engineering.

PROJECTS

Selected projects focusing on control, semantic mapping, and autonomous exploration.

SAGE – Semantic-Aware Guided Exploration with Persistent Memory

Master's Thesis

📅 2025 – 2026

- Developed a real-time semantic exploration system using RGB-D camera streams for depth fusion, point cloud generation, and spatial perception.
- Leveraged 2D navigation SLAM and 3D semantic point cloud maps, combined through geometry-aware fusion pipelines to enhance spatial understanding.
- Integrated image preprocessing, object detection (YOLO-E), and 3D semantic fusion (OpenFusion) to create persistent scene representations for open-vocabulary reasoning.
- Combined multi-source sensor fusion and behavior tree decision modules to enable autonomous object search and exploration under real-time constraints.

Nonlinear Model Predictive Controller (nMPC) for Differential Drive Mobile Robot (DDMR)

Frameworks: ROS 1 & 2 / CasADi / Python / PyTorch / Docker

📅 2024 – 2025

- Designed an nMPC local planner predicting future robot states via kinematic modeling and online optimization.
- Benchmarked against DWA and TEB planners in Gazebo, demonstrating smoother, constraint-compliant trajectories.

Full portfolio: KevinEppacher.github.io

LANGUAGES

English



German



Italian



REFEREES

Alexis Müller

@ Software Team Lead, Baubot GmbH

✉ alexis.mueller@baubot.com

Software engineering reference

Dr. Michael Zillich

@ CTO, Blue Danube Robotics (AIRSKIN)

✉ zillich@airskin.io

Supervised industrial robotics projects